

# One intent. Different computers. Traceable results.

A synthetic 12-guest wedding becomes a complete, inspectable activity.



KEEP FAMILIAR

Keep familiar circles together.

| TABLE 1                                                                         | TABLE 2                                                                         | TABLE 3                                                                         |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <div>Guest 05</div> <div>Guest 06</div> <div>Guest 07</div> <div>Guest 08</div> | <div>Guest 01</div> <div>Guest 02</div> <div>Guest 03</div> <div>Guest 04</div> | <div>Guest 09</div> <div>Guest 10</div> <div>Guest 11</div> <div>Guest 12</div> |

MIX CIRCLES

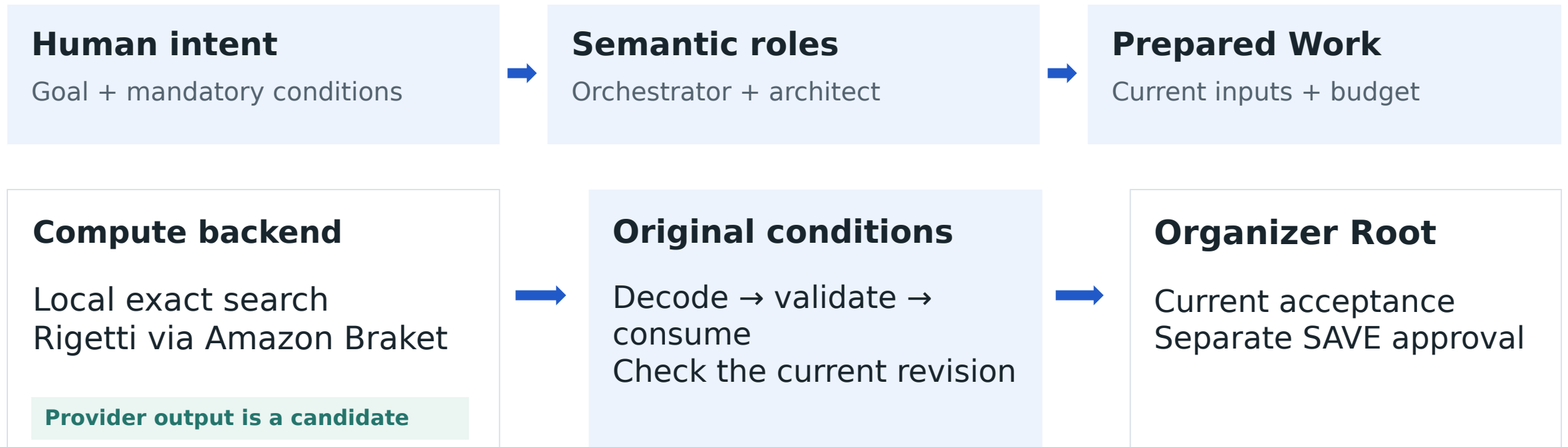
Help different circles meet.

| TABLE 1                                                                         | TABLE 2                                                                         | TABLE 3                                                                         |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <div>Guest 01</div> <div>Guest 02</div> <div>Guest 07</div> <div>Guest 08</div> | <div>Guest 03</div> <div>Guest 04</div> <div>Guest 09</div> <div>Guest 10</div> | <div>Guest 05</div> <div>Guest 06</div> <div>Guest 11</div> <div>Guest 12</div> |

Two measured plans. Same mandatory conditions. Each checked against original conditions and saved.

# The computation changed. The commitment did not.

A request, its conditions and its current owner remain connected to the result.



Time, local memory and evidence follow the activity across these boundaries.

Explanatory view of recorded W2-W4 stages; one observed owner.

# A familiar optimization problem becomes an operating-system case study.

The contribution is how a changing activity stays connected around its solver.

## The established problem

Guests, capacities, relationships and an objective define seating.

## The activity we trace

Intent changes the work.  
A revision changes what is current.  
A measured candidate needs acceptance.

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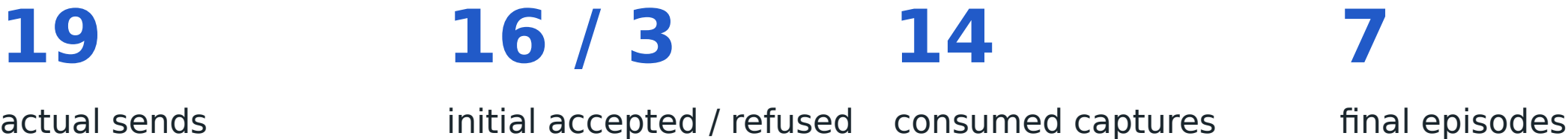
|                  |                               |                 |                            |
|------------------|-------------------------------|-----------------|----------------------------|
| L1               | L2                            | L3              | L4 / L5                    |
| <b>PuLP</b>      | <b>Bellows &amp; Peterson</b> | <b>OR-Tools</b> | <b>Masala + research</b>   |
| Set partitioning | Mixed-integer model           | CP-SAT example  | Interchangeable optimizers |

# Nineteen real model calls shaped seven recorded episodes.

One model: Gemini 2.5 Flash. Two roles: intent orchestrator and requirement architect.



Outlined: 14 captures consumed in the final stories. Amber: initial local acceptance failed.



- 03 Missing required report output.
- 09 Valid null objective met an overstrict consumer.
- 15 A grouped amendment met an overstrict collection bound.

# A different wish selected a different objective and a different local plan.

The model chose between two declared profiles; the solver produced the arrangement.

CALL 05 · EXACT FIELD EXCERPT

"objective\_profile":  
"KEEP\_FAMILIAR\_V01"

CALL 07 · EXACT FIELD EXCERPT

"objective\_profile":  
"MIX\_CIRCLES\_V01"

| TABLE 1                                                                             |          |
|-------------------------------------------------------------------------------------|----------|
|    | Guest 01 |
|    | Guest 02 |
|   | Guest 03 |
|  | Guest 04 |

| TABLE 2                                                                             |          |
|-------------------------------------------------------------------------------------|----------|
|    | Guest 05 |
|    | Guest 06 |
|   | Guest 07 |
|  | Guest 08 |

| TABLE 3                                                                             |          |
|-------------------------------------------------------------------------------------|----------|
|    | Guest 09 |
|    | Guest 10 |
|   | Guest 11 |
|  | Guest 12 |

| TABLE 1                                                                               |          |
|---------------------------------------------------------------------------------------|----------|
|    | Guest 01 |
|    | Guest 02 |
|   | Guest 09 |
|  | Guest 10 |

| TABLE 2                                                                               |          |
|---------------------------------------------------------------------------------------|----------|
|    | Guest 03 |
|    | Guest 04 |
|   | Guest 05 |
|  | Guest 06 |

| TABLE 3                                                                               |          |
|---------------------------------------------------------------------------------------|----------|
|    | Guest 07 |
|    | Guest 08 |
|   | Guest 11 |
|  | Guest 12 |

Both local plans: all original conditions satisfied; objective score 0 in their own profile.

# A changed request created a new condition, with its predecessor intact.

Guests 07 and 08 move together: table 2 → table 1 → verify without another search.



## CALL 17 · EXACT EXCERPT (ONE OF TWO AMENDMENTS)

```
"operation": "REPLACE",  
"subjects": ["guest_07"],  
"tables": ["table_1"],  
"replaces": ["added:cefff2237df9864e94438b6db02b436e"]
```

Old SR1 plan under SR2:  
refused at the domain  
boundary.

Owner input was scripted;  
Root review was real.

# “Check this plan” performed less work than “make a plan”.

A finite capability catalogue turns semantic differences into different execution paths.

## GENERATE / REVISE

**review → compile → solve → validate → consume**

4 searches across A, B, SR1, SR2

## VERIFY / SR3

**validate → consume**

0 search; 0 QUBO; 0 encoding

## AMBIGUOUS

**clarify the objective**

0 native task; no invented plan

```
"task_kind": "VALIDATE_EXISTING",  
"objective_profile": null,  
"needed_capabilities": ["VALIDATE_ORIGINAL"]
```

# Budget and memory affected real work in the local precursor.

Recorded W2 evidence: useful diagnostics and a scoped LocalDRS return path.

## 6 compute units

1 seed + 4 diagnostics + 1 consumer

| Table | Cohesion | Mixing |
|-------|----------|--------|
| 1     | 4        | 2      |
| 2     | 4        | 2      |

## LocalDRS

Record → search → eligibility  
→ current Root descent

A conflict stopped its scoped path.  
Fresh review enabled recovery.

Memory did not restore old rights.

# The numerical adapter changed while the original conditions stayed.

A documented engineering change from an annealing-oriented plan to gate-based hardware.

## Planned route

### D-Wave / annealing

Owner-reported access obstacle from Albania. No D-Wave task was executed in this study.



## Implemented route

### Braket / Rigetti

Exact reduction + OpenQASM  
Two hardware tasks  
Original seating checks retained

**Developers replaced the adapter. The system did not autonomously fail over between QPUs.**

# An exact reduction preserved every feasible seating assignment.

Classical mathematics and preparation remained explicit around the QPU.

36

**binary variables**

12 guests × 3 tables

10

**logical qubits**

5 free pairs × 2 bits

24

**valid assignments**

within 1,024 reduced states

**KEEP:  $E = 25H + F_{\text{keep}}$**

**MIX:  $E = 13H + F_{\text{mix}}$**

H = 0 means all mandatory rules hold.

Pair substitution produces an up-to-4-local cost Hamiltonian, not a 10-variable QUBO.

34,650 balanced arrangements → 24 original feasible

# Two real tasks returned measurements from the same Rigetti QPU.

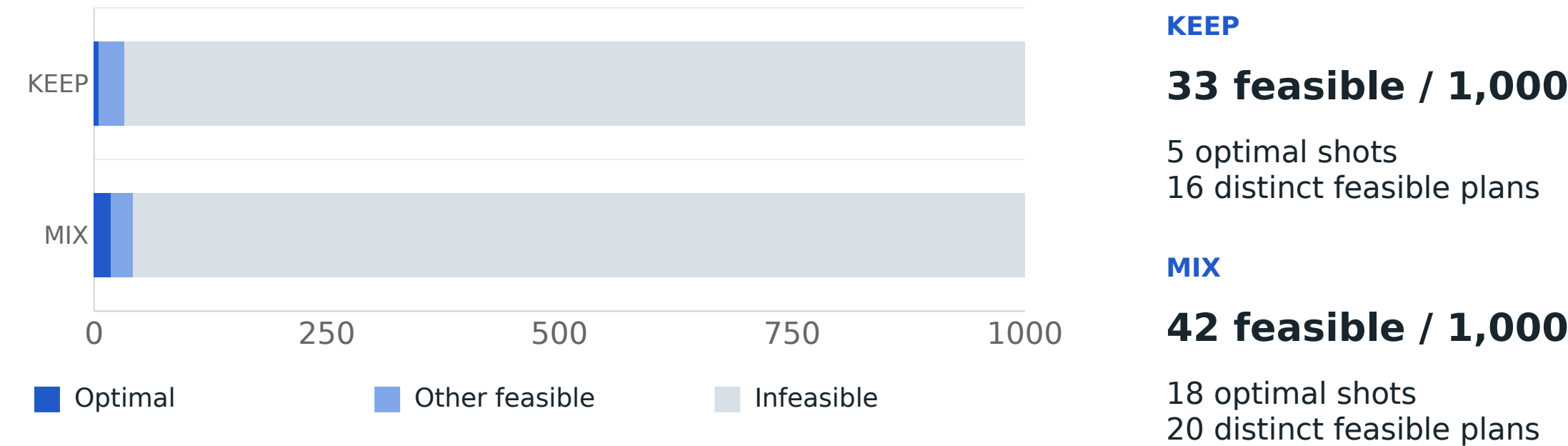
Amazon Braket · Rigetti Cepheus-1-108Q · recorded 24 September 2026.



| Task alias | Provider result | Recorded application result |
|------------|-----------------|-----------------------------|
| KEEP       | COMPLETED       | VALIDATED_AND_SAVED         |
| MIX        | COMPLETED       | VALIDATED_AND_SAVED         |

# Every shot was checked against the original seating conditions.

Feasible samples were a small subset; their selected plans still matched the finite optimum.



Invalid fraction is not a hardware-noise estimate. Distinct plans and shot counts are different quantities.

# A saved plan can be traced to one specific hardware measurement.

No failed sample was replaced by a classical solution.

## MEASURED QUBITS [0...9]

KEEP · raw index 63  
1010000001  
basis = 517

## MEASURED QUBITS [0...9]

MIX · raw index 31  
0010010010  
basis = 292

- 1 Decode** Five pair codes + fixed pair
- 2 Validate** All original conditions; score 0
- 3 Consume** Actual native Work consumes plan
- 4 Accept + save** Current Root; separate JSON/HTML save

# Waiting preserved task identity without extending old authority.

Submission and later result consumption remained separate, bounded decisions.

**07:18 UTC**

**QUEUED**

**09:00 UTC**

**RESUME**

---

Two task IDs created

---

No replacement  
submissions

---

Both tasks completed

---

Read those same task  
IDs

**~102 minutes**

Creation-to-completion: queue + execution.  
Not isolated QPU compute time.

**Read existing task → current review**  
**Prepare fresh consumption → approve**  
**SAVE**

The 96.71 s resume process is not a device  
benchmark.

# Each participant received a different projection of the same activity.

Disclosure controls protected selected synthetic fields, with explicit trust assumptions.

## Local private context

Private fields stay local.  
Owner retains the full instance.

## Semantic projection

Approved synthetic IDs,  
wishes  
and allowed condition  
catalogue.

## Numeric projection

Approved coefficients,  
circuit  
and task/result identities.

Cloud principal: radiolaria-quantum IAM user.

Pseudonyms still disclose relationship structure; hashes do not anonymize data.

# A measured candidate could fail even when its decoding was coherent.

The original domain rules decided admissibility before acceptance.

## KEEP · raw shot 2

00000000100

Decoded, then INVALID  
Table capacities violated  
Guests 01 and 05 must sit apart

## KEEP · raw shot 63

1010000001

Decoded, then VALID  
All original conditions satisfied  
Consumed and separately saved

Additional recorded controls cover changed sources, foreign result binding and pending-task handling.

# The recorded result can be checked without buying another QPU task.

An external pin binds the accepted portable bytes; installed code checks supported relations.

## 1 Use the accepted implementation

fb136f4... on owner main

## 2 Supply the independent package pin

3c924898...33bf1552

## 3 Run the read-only verifier

Canonical result e4a1a644...cad09b0

```
python demo/run_wedding_seating_v01.py verify ...
```

[Full command + independent pin](#)

No new Root, Host, solver, LLM, AWS or QPU execution.

This validates stored event relationships, not a replay of every native schema.

# The comparison concerns evaluated scope, not a solver leaderboard.

Related work already formulates seating, exchanges optimizers and checks feasibility.

| Reference              | What the cited work evaluates                            | Wedding case study                              |
|------------------------|----------------------------------------------------------|-------------------------------------------------|
| PuLP / OR-Tools        | Set partitioning / CP-SAT with prepared inputs           | Changing intent selects performed work          |
| Bellows & Peterson     | 107 guests; best-found plan after 36 h + human revisions | Explicit revision and acceptance history        |
| Nicholas & Mulligan v2 | CFN; classical and D-Wave mappings; 37-person case       | A Rigetti measurement-to-save chain             |
| Masala Core            | Runtime plugins and interchangeable representations      | Ownership, time and evidence around computation |

Not evaluated in a cited work  $\neq$  impossible in that system. No matched benchmark or quantum speedup.

# A 107-guest profile needs new domain engineering, not a new owner principle.

Architectural extension argument. The observed experiment remains 12 guests.

# 12

## observed guests

Three tables of four  
Fixed pairs and 10-qubit encoding

# 107

## future profile

New capacity rules and guest bounds  
New solver / encoding / cost limits

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Keep explicit conditions, source identities, current review and independent validation.  
Rebuild the numerical profile; do not stretch a fixed 12-guest contract.

# Inspect the words. Trace the sample. Verify the result.

The computation changed. The commitment did not.

## 19 real calls

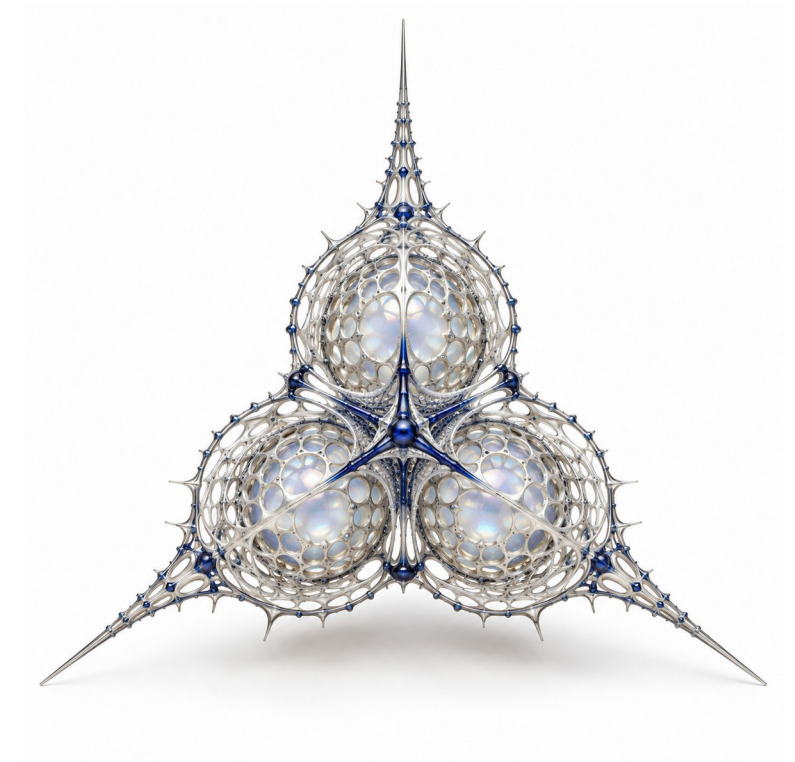
Task choice, objectives and revisions.

## 2 hardware tasks

One measured candidate per saved plan.

## 34 added source paths

Accepted domain extension; common core preserved.



[Technical appendix](#)

[Complete XML Reader](#)

[Sources + scope](#)

Finite case study · No quantum speedup or production certification · Repository access required; new links activate after publication.